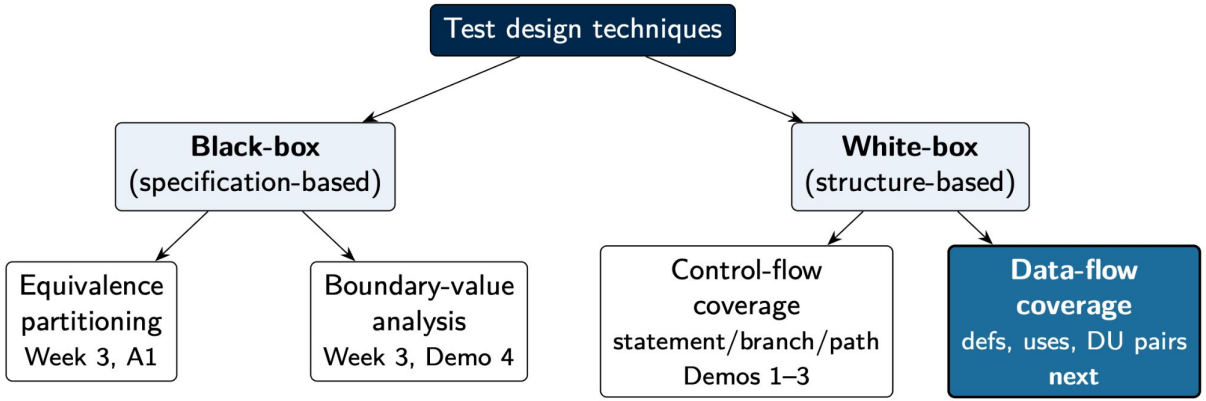


SE 413/513 (Software Testing)

Introduction to Data-Flow Testing

Where Data-flow Testing Fits?



Data Flow Testing

```python
def calculate_price(price, employee):
    discount = 0

    if employee:
        discount = price * 0.10

    discount = price * 0.05

    final_price = price - discount

    return final_price
```

```python
calculate_price(100, True)
```

Suppose a store gives an employee a **10% discount**, and others **5%**.

Can you identify a Bug in the following code?

Data Flow Testing

We can track the

- ***definition (D),*** *and*
- ***use (U)***

```
def calculate_price(price, employee):
    discount = 0                    # D1

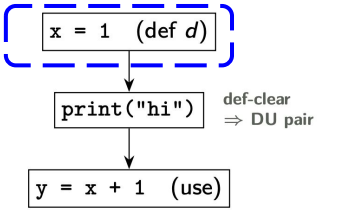
    if employee:
        discount = price * 0.10     # D2

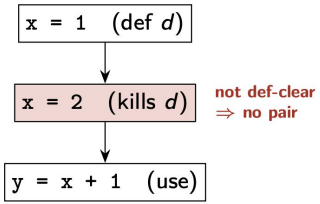
    discount = price * 0.05         # D3  ← redefinition/overwrite

    final_price = price - discount  # U

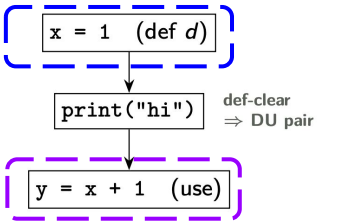
    return final_price # U
```

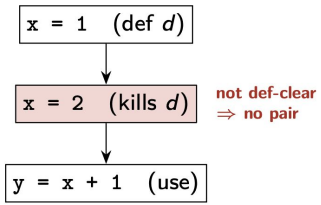
Def-Clear and DU Pairs



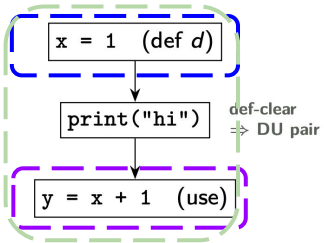


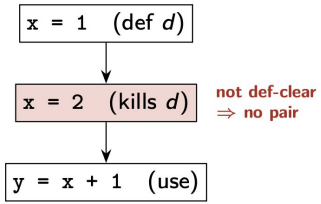
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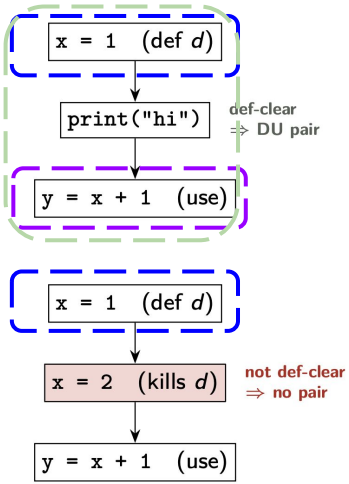


Def-Clear and DU Pairs

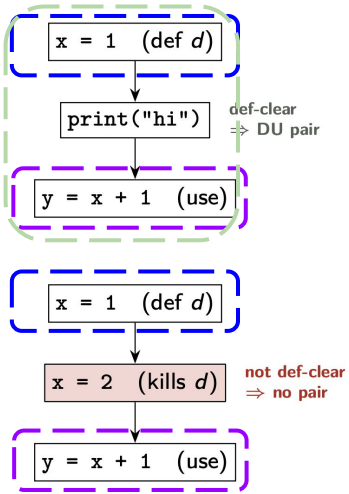




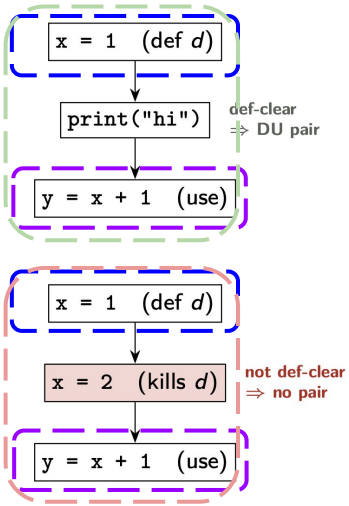
Def-Clear and DU Pairs



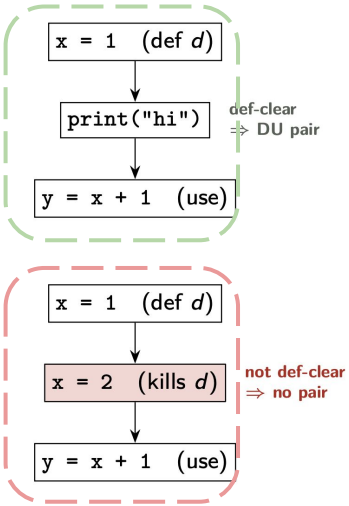
Def-Clear and DU Pairs



Def-Clear and DU Pairs



Def-Clear and DU Pairs



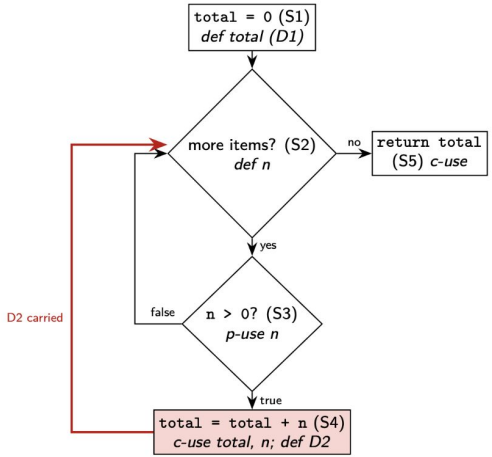
Worked Example: sum_positive()

app/billing/charges.py

```
def sum_positive(numbers):
    total = 0                    # S1
    for n in numbers:            # S2
        if n > 0:                # S3
            total = total + n    # S4
    return total                 # S5
```

Billing reading: charges are positive, credits/refunds are ≤ 0; only charges are summed.

S1	def `total` (**D1**)
S2	def `n`
S3	p-use `n`
S4	c-use `total`, `n`; def `total` (**D2**)
S5	c-use `total`



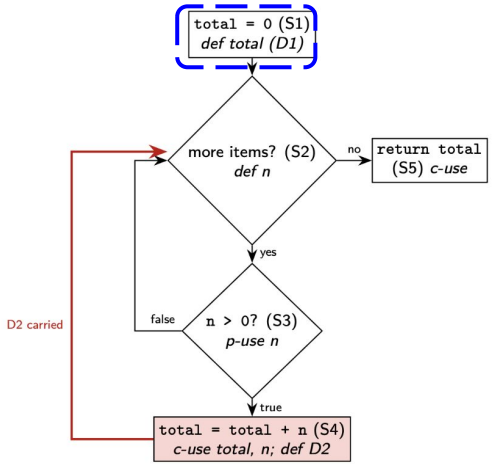
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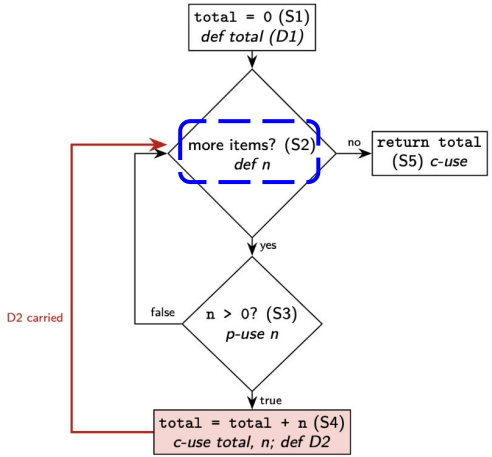
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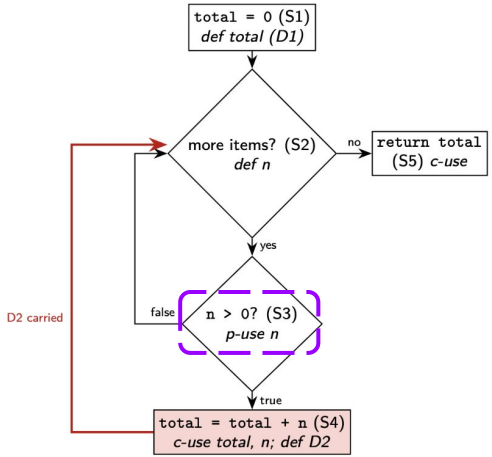
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Billing reading: charges are positive, credits/refunds are ≤ 0; only charges are summed.

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S5	c-use total



p-use (predictive use)

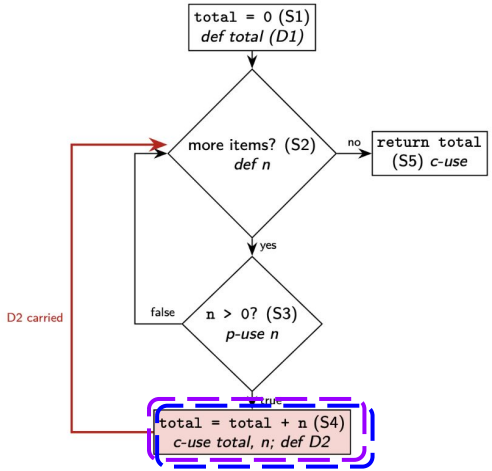
Worked Example: sum_positive()

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```
def sum_positive(numbers):
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S3	p-use n
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S5	c-use total



c-use (computational use)

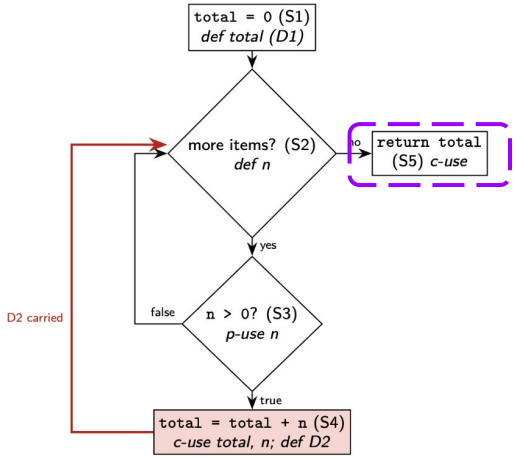
Worked Example: sum_positive()

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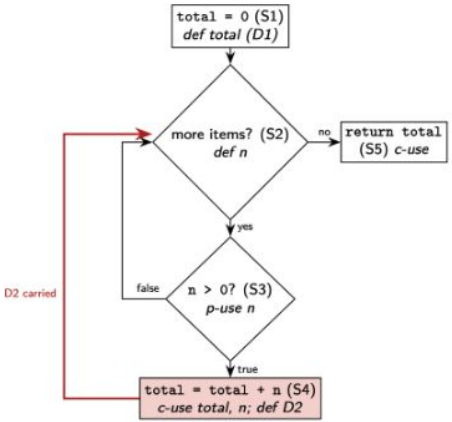


c-use (computational use)

Worked Example: sum_positive()

#	Var	Def	Use	Notes	Covered by (test id)
1	total	D1 (S1)	S4 (c)	first pass through loop	`all-defs:one-charge`
2	total	D1 (S1)	S5 (c)	S4 never runs	`all-defs:credit-only`
3	total	D2 (S4)	S4 (c)	**loop-carried**	`all-uses:two-charges`
4	total	D2 (S4)	S5 (c)	S4 runs, then loop exits	`all-defs:one-charge`
5	n	S2	S3 true (p)	element > 0	`all-defs:one-charge`
6	n	S2	S3 false (p)	element ≤ 0	`all-defs:credit-only`
7	n	S2	S4 (c)	only if S3 is true	`all-defs:one-charge`

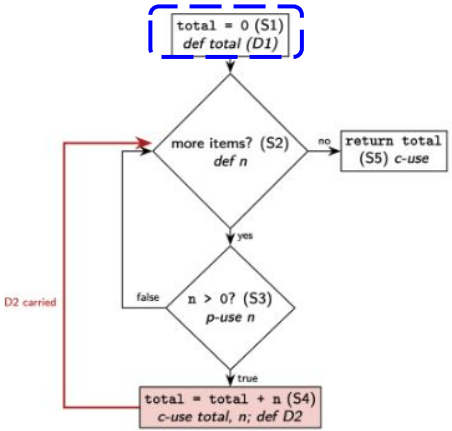
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
+ `[3, -2, 5]`	8	`du-path:loop-carried-via-credit`	a second path for #3



Worked Example: sum_positive()

#	Var	Def	Use	Notes	Covered by (test id)
1	total	D1 (S1)	S4 (c)	first pass through loop	`all-defs:one-charge`
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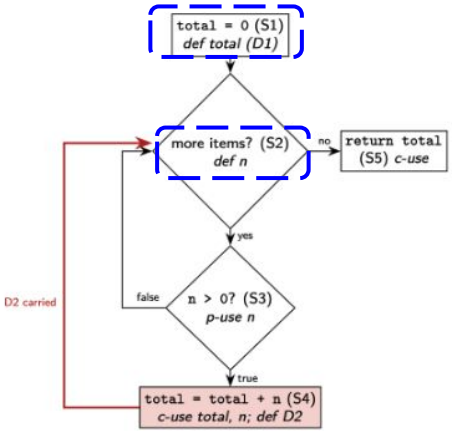
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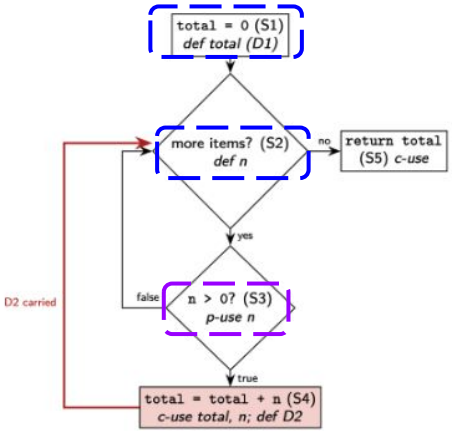
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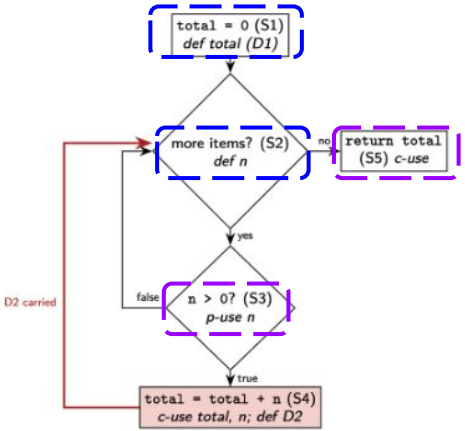
Input	Result	Test id	Criterion reached
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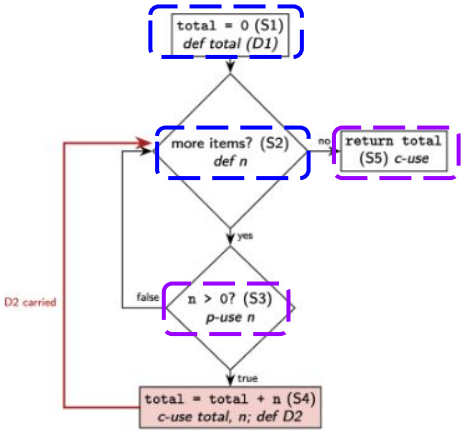
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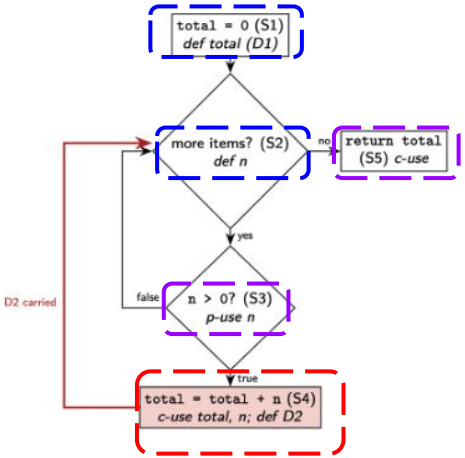
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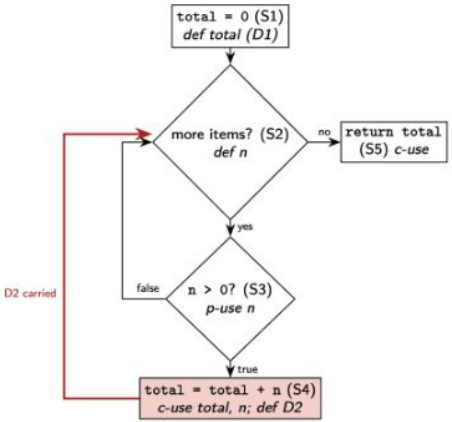
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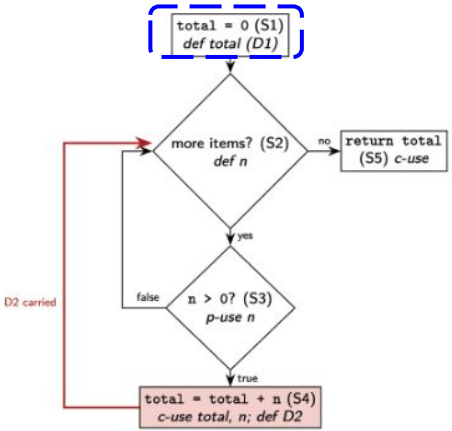
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
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Worked Example: sum_positive()

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7	n	S2	S4 (c)	only if S3 is true	`all-defs:one-charge`

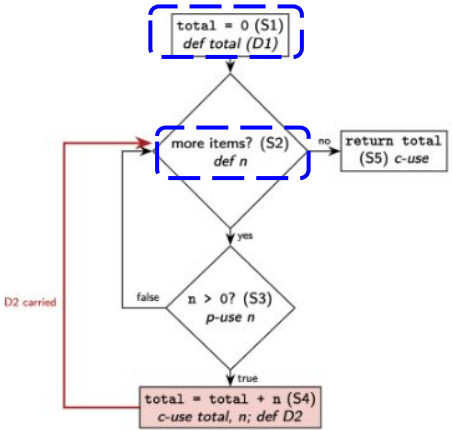
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
+ `[3, -2, 5]`	8	`du-path:loop-carried-via-credit`	a second path for #3



Worked Example: sum_positive()

#	Var	Def	Use	Notes	Covered by (test id)
1	total	D1 (S1)	S4 (c)	first pass through loop	`all-defs:one-charge`
2	total	D1 (S1)	S5 (c)	S4 never runs	`all-defs:credit-only`
3	total	D2 (S4)	S4 (c)	**loop-carried**	`all-uses:two-charges`
4	total	D2 (S4)	S5 (c)	S4 runs, then loop exits	`all-defs:one-charge`
5	n	S2	S3 true (p)	element > 0	`all-defs:one-charge`
6	n	S2	S3 false (p)	element ≤ 0	`all-defs:credit-only`
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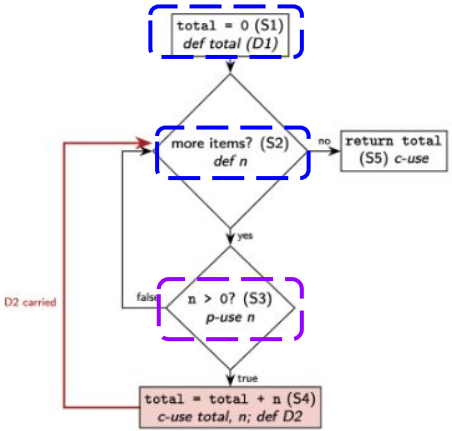
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
+ `[3, -2, 5]`	8	`du-path:loop-carried-via-credit`	a second path for #3



Worked Example: sum_positive()

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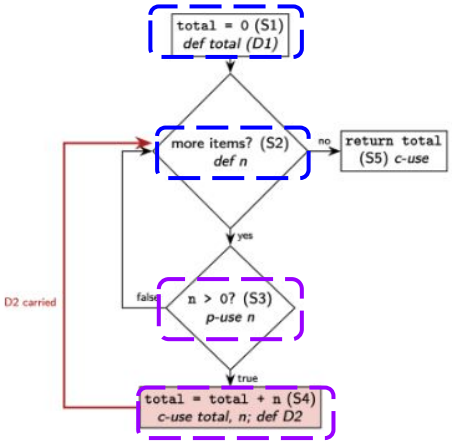
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
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Worked Example: sum_positive()

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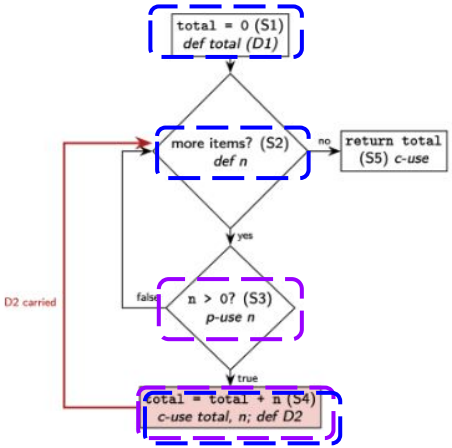
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
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Worked Example: sum_positive()

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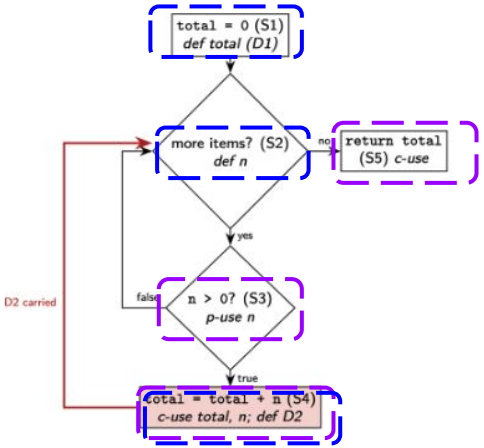
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
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Worked Example: sum_positive()

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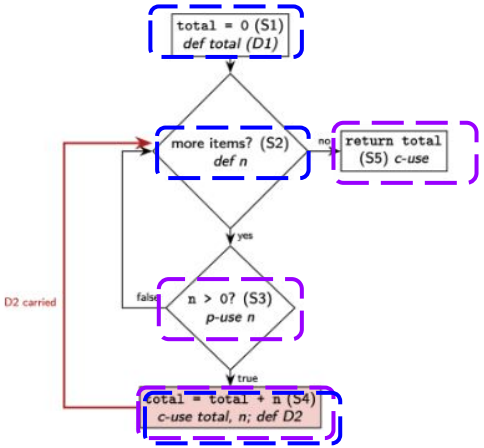
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+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
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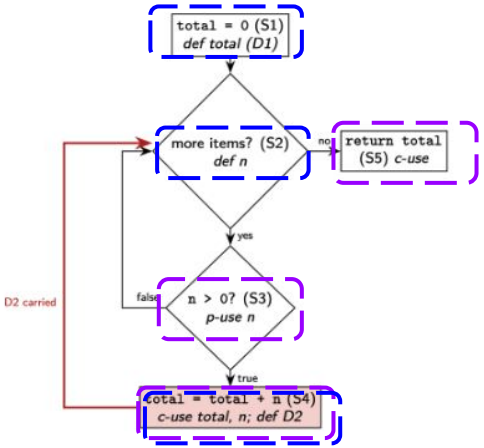


Code demo....

Worked Example: sum_positive()

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1	total	D1 (S1)	S4 (c)	first pass through loop	`all-defs:one-charge`
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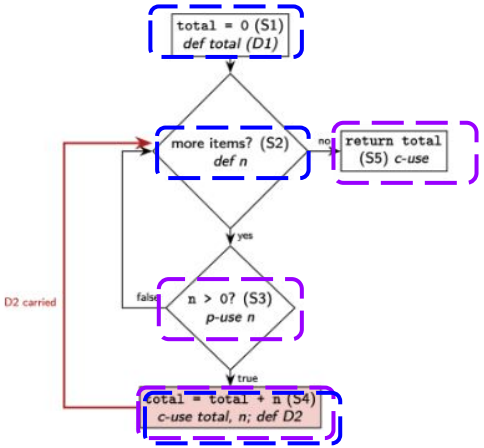
Input	Result	Test id	Criterion reached
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1	total	D1 (S1)	S4 (c)	first pass through loop	`all-defs:one-charge`
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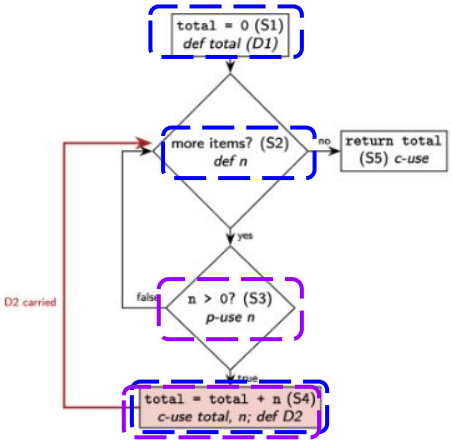
Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
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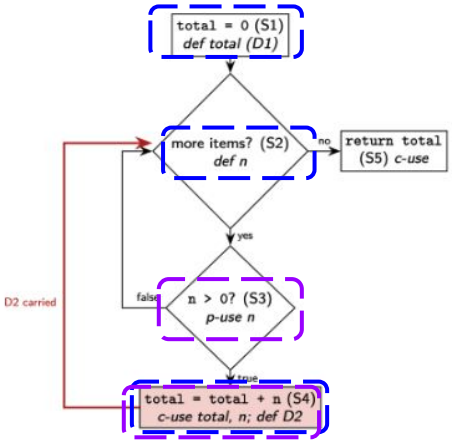
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+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
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Input	Result	Test id	Criterion reached
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Input	Result	Test id	Criterion reached
`[-1], [3]`	0, 3	`all-defs:*`	All-Defs (and 100% branch)
+ `[3, 5]`	8	`all-uses:two-charges`	**All-Uses** (pair #3)
+ `[3, -2, 5]`	8	`du-path:loop-carried-via-credit`	a second path for #3

```
pytest tests/billing/test_charges.py -v \
       --cov=app.billing.charges --cov-branch --cov-report=term-missing
```

```
tests/billing/test_charges.py ....                                       [100%]
Name                       Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------------------
app/billing/charges.py         6      0      4      0   100%
4 passed in 0.11s
```

Predict Before We Run

Comment out the last two cases (`[3, 5]` and `[3, -2, 5]`). Predict: what happens to **branch coverage**?

```
        pytest.param([-1], 0, id="all-defs:credit-only"),
        pytest.param([3], 3, id="all-defs:one-charge"),
        # pytest.param([3, 5], 8, id="all-uses:two-charges"),
        # pytest.param([3, -2, 5], 8, id="du-path:loop-carried-via-credit"),
```

QA